

Cryptosporidium

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**



**World Health
Organization**

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Highlights

- There is strong evidence linking transmission of *Cryptosporidium* with inadequate sanitation, unsafe drinking-water, and poor hygiene. *Cryptosporidium* are major zoonotic waterborne pathogens with many documented drinking-water outbreaks.
- Transmission is primarily by the faecal–oral route. *Cryptosporidium hominis* and *C. parvum* are the major human pathogenic species. Livestock (predominantly cattle and sheep), particularly young animals in the pre-weaning period, are important sources of zoonotic *C. parvum*.
- *Cryptosporidium* infection occurs worldwide; the resulting illness is characteristically prolonged, watery diarrhoea that is usually self-limiting. Malnourished young children and severely immunocompromised individuals may suffer chronic severe diarrhoea with significant loss of fluid and increased morbidity and mortality. *Cryptosporidium* is one of the major causes of morbidity and mortality from diarrhoeal disease in malnourished young children.
- While both *C. hominis* and *C. parvum* cause disease worldwide, there are geographical differences and differences in distribution between urban and rural areas. Seasonal fluctuations have been reported in some countries. The main exposure risks are contact with infected people or animals, ingestion of contaminated food or drinking-water and recreational waters and international travel to areas with inadequate water and sanitation.
- The relative infectivity of *C. hominis* and *C. parvum* is considered to be high but varies between isolates. From the strains examined, calculated ID₅₀s (50% infective doses) for clinical illness ranged from 9 to 1042 oocysts.
- Standard methods are available for detection and quantification of *Cryptosporidium* oocysts in water. However, they are not specific to pathogenic *C. hominis* and *C. parvum*, and they do not indicate viability or infectivity. Animal models and cell culture assays can be used to determine infectivity but are not widely available for environmental samples. Molecular methods are available for identifying species.
- *Cryptosporidium* oocysts are commonly present in sewage, treated wastewater and surface water, and have been detected in vulnerable groundwater and inadequately treated drinking-water. *Cryptosporidium* persists for long periods in water.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from *Cryptosporidium*. Because livestock are a major reservoir for *Cryptosporidium parvum*, a comprehensive approach that also deals with the management of animal faeces is necessary to control transmission.
- The reduction of *Cryptosporidium* oocyst concentrations by wastewater and drinking-water treatment varies depending on the process. Removal is commonly based on physical removal (e.g. size exclusion). *Cryptosporidium* is highly resistant to chemical disinfectants but is sensitive to UV light and ozone.
- Due to the long environmental persistence of *Cryptosporidium* oocysts and their high resistance to some chemical disinfectants such as chlorine, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator for the absence of *Cryptosporidium* in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of *Cryptosporidium*.

1 General description

Cryptosporidium are coccidian protozoan parasites with a complex life cycle including sexual and asexual reproduction (Guérin & Striepen, 2020). Infected hosts shed thick-walled *Cryptosporidium* oocysts with a diameter of 4 µm to 6 µm within the faeces (Betancourt, 2019). When ingested by a new host, the oocyst wall excysts and four motile sporozoites are released and infect the small intestine. The sporozoites transform over several asexual life stages, followed by sexual reproduction to produce thick- and thin-walled oocysts. The thin-walled oocysts can excyst within the same host and start a new life cycle leading to heavy infections, while thick-walled oocysts are excreted in faeces (Leitch & He, 2012; WHO, 2009).

Of the 44 *Cryptosporidium* species, 19 have been reported in humans worldwide; some frequently (*C. hominis*, *C. parvum*) and others occasionally (*C. canis*, *C. felis*, *C. cuniculus*, *C. ubiquitum*, *C. viatorum*) (Ryan et al., 2021). *Cryptosporidium* have been found in faeces of many types of livestock, including cattle, buffalo, sheep, goat, pig, horse, camel, duck, geese, chicken, and others (Ryan et al., 2014; Vermeulen et al., 2017; Betancourt, 2019). *Cryptosporidium* mainly cause intestinal infections which are more prevalent in young animals and these animals shed high concentrations of oocysts in their faeces. While livestock are common sources of zoonotic *C. parvum*, many species of livestock and companion animals can carry non-zoonotic *Cryptosporidium*, which are not relevant for human health.

2 Human health effects

Cryptosporidium infection (cryptosporidiosis) can result in watery diarrhoea of sudden onset, abdominal pain, nausea and vomiting and low-grade fever. These symptoms usually last for 2 weeks but can last for up to 4 weeks and are ultimately self-limiting. Significant weight loss can occur. The symptoms are difficult to control and can lead to profound weight loss, prostration, and increased morbidity and mortality (Chappell et al., 1999; Chappell et al., 2006; Chalmers & Davies, 2010; Betancourt, 2019). Malnourished children under 2 years of age experience moderate to severe diarrhoea that can be persistent and result in increased morbidity and mortality (Kotloff et al., 2013).

Cryptosporidiosis can produce long-term health effects. In malnourished children these include growth faltering and cognitive deficit disorders (Khalil et al., 2018). Severely immunocompromised individuals may suffer chronic severe diarrhoea with significant loss of fluid, as infections spread throughout the gastrointestinal tract or to other sites (Hunter & Nichols, 2002). Meta-analysis of five case control studies in high income countries found that individuals were six times more likely to report chronic diarrhoea and weight loss up to 28 months after a *Cryptosporidium* infection than were controls, and long-term abdominal pain, loss of appetite, fatigue, vomiting, joint pain, headache and eye pain were also between 2 to 3 times more likely (Carter et al., 2020). Seronegative reactive arthritis has been reported in adults and children, including one report of Reiter's syndrome (Cron & Sherry, 1995). It has also been suggested that *Cryptosporidium* infection may cause relapses in Crohn's disease and ulcerative colitis (Manthey et al., 1997). There is no effective therapy or vaccine for cryptosporidiosis (Checkley et al., 2015).

3 Epidemiology

Cryptosporidiosis is a very common gastrointestinal disease of humans. While both *C. hominis* and *C. parvum* cause disease worldwide, there are geographic differences. *Cryptosporidium hominis* is dominant in large parts of the world, including low-income countries, but in Europe and New Zealand, both *C. parvum* and *C. hominis* are prevalent, while *C. parvum* is dominant in the Middle East (Ryan et al., 2014). There are also differences in distribution between urban and rural areas, with *C. hominis* more common in urban areas and *C. parvum* more common in rural areas. People more likely to become infected include children under 5 years of age, caregivers of infected individuals including parents, travellers to countries where infection is endemic and individuals who handle infected livestock (Ryan et al., 2014; Painter et al., 2016). *Cryptosporidium* are among the most important causes of diarrhoeal disease and deaths among children under 5 years of age, especially in low-income settings (Kotloff et al., 2013; GBD, 2020; Khalil et al., 2018). In high-income countries, reported incidence rates of 4.4 per 100 000 for the general population, while in children, the incidence rates are higher at 15.8 per 100 000 individuals (ECDC, 2021). The reported rates are believed to be a fraction of the true rates. Cacciò & Chalmers (2016) estimated that actual incidence rates of symptomatic cryptosporidiosis in Europe are at least approximately 100-fold higher.

Asymptomatic carriage rates of 1.3% to 3.8% have been found among immunocompetent children in childcare settings in France, Spain, the United Kingdom (UK), and the United States of America (USA) (Cordell & Addiss, 1994; Davies et al., 2009). In Peru, 18% of asymptomatic carriage was reported among young children. It was found to be associated with poor growth (Checkley et al., 1998), indicating that asymptomatic infection may also cause an important burden of disease.

Seasonal fluctuations have been reported in some countries. *Cryptosporidium hominis* is more prominent in autumn in the UK, the Netherlands and New Zealand, and *C. parvum* is more prominent in spring in Canada, Ireland, New Zealand and the Netherlands (Wielinga et al., 2008; Lake et al., 2008; Ryan et al., 2014). This could reflect the sources of infective oocysts. Springtime peaks tend to be linked to *C. parvum* from livestock, and summer-autumn peaks to *C. hominis* from travel abroad and recreational water exposure (Lake et al., 2008; Zintl et al., 2009; Chalmers, 2012; Gharpure et al., 2019).

4 Transmission

Cryptosporidium are transmitted via the faecal-oral route. In lower-and middle-income countries, crowded living conditions, animal contact and open defecation are responsible for the majority of *Cryptosporidium* cases (Bouzid et al., 2018). The vast majority of cases of cryptosporidiosis occur sporadically, but common source outbreaks can occur (Painter et al., 2016). In higher-income countries, outbreaks have been reported via ingestion of contaminated water (particularly drinking-water, swimming pool water, at spray parks and natural recreational waters), food, and also contact with infected humans and animals (Karanis et al., 2007; Baldursson & Karanis, 2011; Robertson & Chalmers, 2013; Cacciò & Chalmers, 2016; Gharpure et al., 2019). Globally, 239 waterborne outbreaks were reported between 2011 and 2016 (Efstratiou et al., 2017). In the USA, an overview of 444 cryptosporidiosis outbreaks (Gharpure et al., 2019) from 2009 to 2017 identified the most common modes of transmission:

waterborne (41.2%), person-to-person (19.8%), animal contact (19.4%), and foodborne (5.0%). Most waterborne outbreaks ($n = 183$) were associated with recreational waters (92.9%), followed by drinking-water (4.4%). In England and Wales, 178 outbreaks were reported between 2009 and 2017. Of these, 46% were associated with recreational water, 42% with animal contact, 2% each with person-to-person contact and food, and 1% with drinking-water (Chalmers et al., 2019). Calves and lambs are important sources of zoonotic *C. parvum*, as indicated by frequent reports of zoonotic transfer (Chalmers & Giles, 2010), which has implications for waterborne transmission (Lake et al., 2007; Pollock et al., 2009).

In general, the relative infectivity of *C. hominis* and *C. parvum* is considered to be high.¹ For both species, a series of challenge studies in healthy adult volunteers has been undertaken from which ID₅₀s (50% infective dose, i.e. number of organisms required to infect 50% of exposed subjects) have been calculated. For *C. hominis*, the ID₅₀ was 10 oocysts for diarrhoeal illness and 83 oocysts for infection based on faecal samples positive for oocysts. (Chappell et al., 2006). For *C. parvum* there was considerable variability in infectivity between isolates with ID₅₀s ranging from 12 (TAMU), 132 to 165 (IOWA), 179 (UCP) to 455 (Moredun) (DuPont et al., 1995; Okhuysen et al., 1999; Okhuysen et al., 2002; Enger, 2013). The ID₅₀s for clinical outcomes for three different strains of *C. parvum* ranged from 9 to 1042 oocysts (Okhuysen et al., 1999). Many quantitative microbial risk assessments have used the infectivity of the Iowa isolate since this was the first dose-response experiment reported. Later studies have amalgamated the dose-response data of the different isolates to develop dose-response models that incorporate isolate variability (Teunis et al., 2002; WHO, 2009; Messner & Berger, 2016). One study identified a significant chance of infection with a single oocyst (Messner et al., 2001).

Symptoms of infection can manifest between 3 and 14 days (typically 5 to 7 days) after ingestion of *Cryptosporidium* oocysts; however, oocysts are detectable in stool samples 1 to 3 days before clinical symptoms begin (Chappell et al., 1999; Chalmers & Davies, 2010). At the peak of infection, humans shed up to 10^5 to 10^7 oocysts/g of faeces (Chappell et al., 1999). Asymptomatic individuals can shed infective oocysts for weeks to months after infection (Jokipii & Jokipii, 1986). Pre-existing anti-cryptosporidial antibodies in individuals do not confer immunity from infection but can protect against diarrheal symptoms (Borad & Ward, 2010). In addition, there is some suggestion that low dose repeat exposures are protective of severe disease (Frost et al., 2005).

5 Detection methods

Standard methods to detect and quantify *Cryptosporidium* oocysts in water are available (e.g. ISO, 2006; USEPA, 2012). Oocysts are concentrated using high volume filtration and elution, concentration by centrifugation, immunomagnetic separation, and enumeration using immunofluorescence microscopy. The methods do not determine the viability or infectivity of the oocysts, and some antibodies used as the fluorescent marker are not specific to *C. hominis* or *C. parvum* and cross-react with other *Cryptosporidium* species. The recovery efficiency of

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be $1-10^2$ organisms or particles, Moderate 10^2-10^4 and Low $> 10^4$.

the detection methods is relatively low, so the actual concentrations are generally higher. However, commercial products, such as uniquely labelled fluorescent oocysts, are available to allow recoveries to be determined in each sample. Molecular methods are available to identify any *Cryptosporidium* species, as well as methods specifically for *C. hominis* and *C. parvum*. These methods do not discriminate between infectious and non-infectious oocysts, but animal models or cell cultures can be used to determine infectivity (Slifko et al., 1997; Schets et al., 2005; Swaffer et al., 2018; Betancourt, 2019).

6 Source and occurrence

Various environmental matrices have been examined for *Cryptosporidium*. A summary of their sources, concentrations and persistence is provided in Table 1 and the subsequent text.

Table 1. Summary of *Cryptosporidium* occurrence data (adapted from Nasser, 2016; Betancourt, 2019)

Source	Prevalence		Concentrations (oocysts/L)		Method
	Studies	% Positive Samples	Studies	Maximums ^a	
Raw Sewage	14	6–100	4 3 3 4	10 ² 10 ³ 10 ⁴ >10 ⁴	(IMS ^b)-IFA ^c
Wastewater sludge	3	38–100	3	10 ³	(IMS-)-IFA
Treated wastewater ^d	20	100	15 4 1	10 ¹ 10 ² 10 ³	(IMS-)-IFA
Surface water (rivers, lakes, streams)	34	5–100	13 8 10 4 4 4	10 ⁰ 10 ¹ 10 ² 10 ³ >10 ³ NR ^e	(IMS-)-IFA
Seawater	3	NR	1 1 1	10 ⁻² 10	(IMS-)-IFA
Groundwater (wells, boreholes, springs)	10	5–67	2 2 4 2	10 ⁻¹ 10 ⁰ 10 ¹ NR	(IMS-)-IFA
Drinking-water	16	1.1–40	3 0 9 4	10 ⁻² 10 ⁻¹ 10 ⁰ >10 ⁰	(IMS-)-IFA

^a All maximum values are single values except treated wastewater which are maximum averages, concentrations for sewage sludge are /g dry weight.

^b immunomagnetic separation to purify *Cryptosporidium*

^c immunofluorescence assay to detect and enumerate *Cryptosporidium*

^d Only studies positive for oocysts were reported.

^e Not reported

6.1 Faecal pollution sources

High prevalences of *Cryptosporidium* in raw sewage, sewage sludge, septage (septic tank contents), and faecal sludge (pit latrine contents) are reported in both high- and low-income countries (Table 1). The average prevalence from 14 studies of raw sewage was 61%, with concentrations ranging from less than 1 to 60 000 oocysts/L. *Cryptosporidium* can generally be found in raw sewage and faecally contaminated surface waters year-round. The concentration of *Cryptosporidium* in raw sewage and faecally contaminated water varies between studies, probably due to the composition of the sewage, type of water, assay efficiency, seasonality, geographic location and prevalence of *Cryptosporidium* shedders in the population (Jagai et al., 2009; Montemayor et al., 2009; Koompapong & Sukthana, 2012). Cattle produce the highest loads of *Cryptosporidium* in the environment, but human faecal discharges were the dominant source of *Cryptosporidium* in rivers in most regions (Vermeulen et al., 2019).

6.2 Surface water

Cryptosporidium have been detected in all types of surface water although the prevalence and concentration can vary greatly (Table 1). Combining 38 studies from 16 countries (n = 9384) the mean percentage of samples that contained *Cryptosporidium* was 47% (WHO, 2009). In pristine sites, the geometric mean concentrations were between 0.001 and 0.01 oocysts/L, while maximum concentrations per site were approximately 1/L. In heavily contaminated sites the geometric mean concentrations were approximately 10 to 100 oocysts/L, with maxima between 100 and 1000 oocysts/L. Most studies reported oocyst concentrations without correction for the recovery efficiency of the detection method. An overview of surface water in the United States of America (USA) showed that *Cryptosporidium* were found in 9.1% to 100% of samples with geometric mean concentrations ranging from 0.003 to 1920 oocysts/L (Rose et al., 1997). More recent data on *Cryptosporidium* occurrence and concentrations in surface water are summarized by Betancourt (2019), with widely variable occurrences of 4.5% to 100% positive samples and average concentrations ranging from 0.04 to 1500 oocysts/L among different fresh and marine surface water sources, geographic locations and water uses.

Infective *Cryptosporidium* oocysts can persist for long periods in water with survival ranging from weeks to months depending on temperature (King & Monis, 2007).² At temperatures below 15 °C, oocysts can maintain high levels of (mouse) infectivity for periods of at least 24 weeks (Fayer et al., 1998). At temperatures of 20 °C and 25 °C, some survival was observed by mouse infectivity after 24 to 12 weeks respectively (Fayer et al., 1998). King et al. (2005) observed complete inactivation of 5×10^4 oocysts/mL in de-ionized water at 20 °C and 25 °C after 12 and 8 weeks respectively using cell culture. At 30 °C and 37 °C, complete inactivation occurred within 500 hours and 72 hours respectively. Freeze-thaw cycles may inactivate oocysts (Robertson & Gjerde, 2004). *Cryptosporidium* oocysts are reduced by exposure to sunlight in both tap and surface water (King et al., 2008). At higher sunlight exposures of 10 000 kJ/m² to 30 000 kJ/m² infectivity reductions were greater than 2 log₁₀ to 3 log₁₀, depending on water type. At higher DOC levels, inactivation was slower (King et al., 2008). In another study on the die-

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

off of *C. parvum* in tap water, exposure to summer sunlight at 33 °C resulted in a loss of only 1.58 log₁₀ after 7 days (Nasser et al., 2007).

6.3 Groundwater

Cryptosporidium are more commonly associated with contaminated surface water, but can be detected in vulnerable groundwater, i.e. impacted by surface water (Table 1). A study in Bulgaria found *Cryptosporidium* oocysts in 67% of wells and springs, with an average concentration of 0.02 oocysts/L (Karanis et al., 2006). In Portugal, oocysts were found in 59% of untreated groundwaters (Lobo et al., 2009). In the USA, oocysts were found in 11% of groundwaters sampled, including 5% of the vertical wells, 20% of the springs, 50% of the infiltration galleries and 45% of the horizontal wells (Hancock et al., 1998). Average concentrations were less than 0.01 to 5 oocysts/L (mean of 0.19 oocysts/L). Similar results were found in studies in Canada, Germany, Norway, Nepal, and India (Daniels et al., 2018; Betancourt, 2019).

6.4 Drinking-water

Cryptosporidium oocysts are typically not present or only present at low concentrations (less than 1 oocyst/L) in drinking-water produced by well-designed and well-operated conventional drinking-water treatment plants (including coagulation, flocculation, sedimentation and filtration). Studies of drinking-waters that were vulnerable to contamination have detected *Cryptosporidium* (Table 1) with occurrence rates (as % positive samples) between 1.1% and 40% and concentrations typically less than 1 oocyst/L. In the USA two large studies detected *Cryptosporidium* in 7% to 14% of drinking-water samples at average concentrations of 0.06 to 0.068 oocysts/L (WHO 2009). In the UK, a decade-long monitoring program in potentially vulnerable drinking-waters revealed that *Cryptosporidium* oocysts were occasionally present in treated drinking-water. During the period 2000 to 2002, a total of 97 999 samples were analysed (total volume = 115 303 050 L) with 5.5% positive, and an average oocyst concentration of 0.0002 oocysts/L (Smeets et al., 2007). From 2000 to 2004 a total of 201 136 samples were collected with 3.0% containing *Cryptosporidium* oocysts with a maximum concentration of 0.49 oocysts/L (WHO, 2009). Improvements in treatment and abandonment of high-risk supplies saw the percentage of positive samples fall from 8 to 1%. Most of the available data are from high-income countries; data from low- and middle-income countries are limited. However, *Cryptosporidium* oocysts were detected in 10% of treated drinking-water samples from the Philippines (1/10 samples), with a concentration of 0.06 oocyst/L, and in 27% of untreated water samples from Thailand (25/93 samples), 39% of samples from Malaysia (17/44 samples), and 48% of samples from the Philippines (11/23 samples), with concentrations ranging from 0.13 oocysts/L to 0.57 oocysts/L (Kumar et al., 2016). In household taps served by an intermittent water supply in Nagpur India, an increase in genetic markers were detected for all waterborne pathogens analysed compared to those served by a continuous water supply, except for *Cryptosporidium* (Bivins et al., (2021). In this study, a single sample from both the intermittent water supply and continuous water supply taps, identified *Cryptosporidium* RNA gene target.

6.5 Soil and crops

Cryptosporidium have been found in the topsoil of dairy farms in a watershed in the USA (Barwick et al., 2003). The likelihood of *Cryptosporidium* presence depended on land-use and

soil pH. Dado et al. (2012) found *Cryptosporidium* in 6% of soils of public parks in Spain. A study in Korea found *Cryptosporidium* by molecular methods in 33% of crop soils with animal husbandry farms as a suspected source (Hong et al., 2014). A systematic review and meta-analysis of the global prevalence of intestinal protozoan contamination of vegetables and fruits noted that 11% of samples of raw vegetables contained *Cryptosporidium* (Badri et al., 2022).

6.6 Surfaces

Limited data are available on the presence or survival of *Cryptosporidium* on surfaces. Desiccation killed 97% of oocysts (starting concentrations of 10^6 oocysts/mL) within 2 hours at room temperature (Robertson et al., 1992). Similar rates (95% in 4 hours at room temperature) were noted by Deng & Cliver (1999). In Lusaka Zambia, *Cryptosporidium* was detected by molecular analysis in a drinking cup, with 1 of 13 samples positive (Asada et al., 2022).

6.7 Seawater

Low concentrations of *Cryptosporidium* oocysts (ranging from 10^{-3} to 10^{-1} /L) have been detected in marine recreational bathing waters affected by wastewater discharges in three tropical settings (Betancourt, 2019). In the experiment by Nasser et al. (2007) noted above, exposure of *C. parvum* in seawater to summer sunlight at 33 °C yielded a loss in infectivity of 3.08 log₁₀ after 7 days.

7 Significance of sanitation

Safe sanitation systems are an important barrier against the spread of *Cryptosporidium* (Betancourt, 2019). Sewage has been identified as a source of a number of drinking-water outbreaks (Wallis et al., 2007; Chalmers, 2012; Guzman-Herrador et al., 2015). However, because animals are a major reservoir for *Cryptosporidium parvum*, a comprehensive approach that also deals with the management of animal faeces is necessary to control transmission (WHO, 2009; WHO, 2022).

8 Significance in drinking-water

Cryptosporidium are major zoonotic waterborne pathogens. There is strong evidence that *Cryptosporidium* are transmitted through drinking-water with a large number of documented outbreaks, including those where sewage was the source of contamination (MacKenzie et al., 1994; Karanis et al., 2007; Baldursson & Karanis, 2011; Chalmers, 2012; Guzman-Herrador et al., 2015; Cacciò & Chalmers, 2016; Efstratiou et al., 2017; Gharpure et al., 2019). The most significant drinking-water outbreak of cryptosporidiosis occurred in Milwaukee, USA in 1993. The outbreak, caused by a failure at a drinking-water treatment plant, resulted in an estimated 403 000 cases and 69 deaths (MacKenzie et al., 1994; Hoxie et al., 1997). Wastewater was identified as the source of the infectious oocysts (Chalmers, 2012). The most common deficiencies associated with outbreaks of cryptosporidiosis in the USA were insufficient treatment or treatment malfunction, untreated and vulnerable groundwater sources, and distribution system deficiencies. In a significant number of outbreaks of cryptosporidiosis, the implicated drinking-water did not contain *E. coli* (Craun et al., 1998). Similar deficiencies have been identified as the cause of drinking-water outbreaks of cryptosporidiosis in other parts of

the world (Baldursson & Karanis, 2011; Efstratiou et al., 2017). Using a fault tree analysis, chronic filtration failures or livestock and rainfall in the catchment were identified as key faults in protozoal (mainly *Cryptosporidium*) outbreaks in the European Union 1995–2005 (Risebø et al., 2007).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from *Cryptosporidium*.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on physical removal (e.g. size exclusion by filtration through media or soil in septic tank drainage areas) or settling in wastewater treatment processes will reduce concentrations of protozoa such as *Cryptosporidium*. Oxidizing chemical disinfectants, such as chlorine, are ineffective against *Cryptosporidium* at the doses and contact times applied to drinking-water, but UV light and ozone are effective at inactivating *Cryptosporidium*.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge wastewater and faecal sludges. In addition, as *Cryptosporidium parvum* also originates from animal faeces, such as livestock, it is necessary to keep these wastes separated from water and from surfaces to which humans are exposed. Removal or inactivation of *Cryptosporidium* by wastewater treatment processes is variable and is largely based on size exclusion (Betancourt, 2019).

There are little data on removal of *Cryptosporidium* by water-based on-site sanitation systems (Betancourt, 2019). However, concentrations of *Cryptosporidium* can be significantly reduced by dry on-site systems when conditions are favourable (Betancourt, 2019). Dry desiccating toilets were better at reducing *Cryptosporidium* oocyst concentrations than composting toilets (Graham et al., 2003).

In centralized sewage treatment plants primary treatment provides limited removal (less than 90%) of *Cryptosporidium* from sewage with most of the oocysts removed present in the sludge (Whitmore & Robertson, 1995; Carraro et al., 2000; Omarova et al., 2018; Payment et al., 2001; Ramo et al. 2017). There is wide variation in oocyst log₁₀ removal efficiencies for secondary treatment technologies: 0.4 to 3.2 log₁₀ for wetlands; 0.2 to greater than 3 log₁₀ for stabilization ponds; 0.4 to 2.2 log₁₀ for activated sludge and trickling filters (Betancourt, 2019; Nasser, 2016; Stadterman et al., 1995; Rodriguez-Manzano et al., 2012; Schmitz et al., 2018). Even after sewage treatment incorporating dual media or sand filtration, *Cryptosporidium* oocysts are frequently detected in treated effluent. For example, one study detected oocysts in 50% of reclaimed water samples used for irrigation (Quintero-Betancourt et al., 2003).

Cryptosporidium oocysts have a high affinity for adsorbing to sludge in biological reactors; more than 50% of oocysts in anaerobic reactors were associated with solids rather than the liquid phase (Kinyua et al., 2016). Because a significant fraction of potentially viable

Cryptosporidium oocysts is retained in sewage sludge, faecal sludge, and septage, special attention must be given to solids management at on-site and centralized facilities. Thermal composting or anaerobic digestion at temperatures between 30 °C and 50 °C can significantly reduce the viability of oocysts, potentially providing as high as 5 log₁₀ reduction in 3 days at 50 °C or in 82 days at 30 °C. The concentration of microbially produced ammonia and other metabolites can influence these rates of decay (Jenkins et al., 1999). *Cryptosporidium* may be resistant to alkaline treatment alone, unless ammonia is present. For example, lime treatment (80 g/kg total solids, pH 12) for up to 72 hours revealed no additional loss of viability, and increased oocyst infectivity by greater than 4-fold (Bean et al., 2007).

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023), control measures that can be applied to manage potential risks from *Cryptosporidium* oocysts include protection of source waters from contamination by human and animal faecal waste (including extra vigilance during heavy rains), application of adequate treatment processes and protection of drinking-water quality during storage and distribution.

Filtration processes are important barriers for *Cryptosporidium* in drinking-water treatment (LeChevallier & Au, 2004; USEPA, 2006; Health Canada, 2012). Full scale conventional treatment with coagulation, floc removal and rapid granular filtration has been determined to remove 2.5 to 3.0 log₁₀ of *Cryptosporidium* oocysts (USEPA, 2006; Health Canada, 2012). A review of published data estimated that conventional treatment, including coagulation, could provide 3.2 log₁₀ removal (range 1.4 to 5.5 log₁₀) while a study of 8 plants in the UK showed removal of 1.4 to 5.4 log₁₀ of *Cryptosporidium* oocysts (Smeets et al., 2007). Other filtration processes, such as slow sand filtration and diatomaceous earth filtration, give removal efficiencies similar to conventional treatment. Membrane filtration can provide removals of more than 4 log₁₀, provided the integrity of the complete membrane system is well-maintained. Passage of water through sandy soils is also very effective in removing oocysts (Hijnen et al., 2005).

Cryptosporidium oocysts are highly resistant to chlorine.³ For a 2 log₁₀ inactivation at pH 7 and 25 °C, the Ct (product of disinfectant concentration and contact time) value for *C. parvum* oocysts was calculated to be 7200 mg.min/L (Korich et al., 1990). More recently Ct values as high as 11 100 mg.min/L in swimming pool water were reported by Shields et al. (2008). Other oxidant disinfectants, such as ozone and chlorine dioxide, differ in effectiveness at inactivating *Cryptosporidium* oocysts. Ozone is effective for inactivation of *Cryptosporidium* oocysts, with Ct values for greater than 2 log₁₀ inactivation ranging from 3.5 mg.min/L to 10 mg.min/L at room temperatures (Finch et al., 1993). Using *C. parvum* inactivation data from Li et al. (2001), Morrison et al. (2022) estimated Ct values for 2 log₁₀ inactivation of *Cryptosporidium* oocysts

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

by ozone to be 4.8 mg.min/L to 8.8 mg.min/L at temperatures of 20 °C to 25 °C. Chlorine dioxide is not effective for inactivating *C. parvum* oocysts in drinking-water. Murphy et al. (2014) estimated a Ct value of 1200 mg.min/L for a 3 log₁₀ inactivation and Chauret et al. (2001) reported Ct values for 2 log₁₀ inactivation of 75, 550, and 1000 mg.min/L respectively for three different sources of *C. parvum*. *Cryptosporidium* oocysts are sensitive to UV light, with 2 log₁₀ inactivation at doses of 5.8 mJ/cm² (Hijnen & Medema, 2005; Hijnen et al., 2006).

Due to the long environmental persistence of *Cryptosporidium* oocysts and their high resistance to some chemical disinfectants such as chlorine, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of *Cryptosporidium* in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of *Cryptosporidium*.

10 ICD-11 classification

XN8LE	<i>Cryptosporidium</i>
XN4ZT	<i>Cryptosporidium hominis</i>
XN9BP	<i>Cryptosporidium parvum</i>
1A32	Cryptosporidiosis

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and the WHO Guidelines on sanitation and health

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.